

LAB NOC 05

Cisco Catalyst 2960 — VLAN Trunk Misconfiguration

Real Hardware | ES / EN | NOC Portfolio

INFORMACION DEL LABORATORIO / LAB INFORMATION

Hardware / Hardware	Cisco Catalyst 2960 (WS-C2960-24TT-L)	Cisco Catalyst 2960 (WS-C2960-24TT-L)
Ticket ID	NOC-LAB-05	NOC-LAB-05
Categoría / Category	VLAN Trunk — Misconfiguration	VLAN Trunk — Misconfiguration
Nivel / Level	CCNA — VLANs / Trunking	CCNA — VLANs / Trunking
Duración / Duration	Aprox. 45 minutos	Approx. 45 minutes
Entorno / Environment	Hardware real (equipo usado)	Real hardware (used equipment)
Relevancia NOC	Diagnostico de conectividad por VLAN, trunk allowed list	VLAN connectivity diagnosis, trunk allowed list

OBJETIVO / OBJECTIVE

OBJETIVO	OBJECTIVE
Diagnosticar y resolver una misconfiguración de trunk VLAN en un Cisco Catalyst 2960: identificar por qué usuarios de VLAN 20 perdieron conectividad, localizar el mismatch en la allowed VLAN list mediante show interfaces trunk, y restablecer el servicio con la configuración correcta.	Diagnose and resolve a VLAN trunk misconfiguration on a Cisco Catalyst 2960: identify why VLAN 20 users lost connectivity, locate the mismatch in the allowed VLAN list using show interfaces trunk, and restore service with the correct configuration.

INCIDENT TICKET — NOC-LAB-05

CAMPO / FIELD	DETALLE / DETAIL
Ticket ID	NOC-LAB-05
Priority / Prioridad	Medium
Service / Servicio	LAN Switching — VLAN Trunking
Device / Equipo	Cisco Catalyst 2960 — Access Layer Switch
Problem / Problema	Usuarios en VLAN 20 sin conectividad. Usuarios en VLAN 10 funcionando normalmente. / Users in VLAN 20 without connectivity. Users in VLAN 10 working normally.
Topology / Topología	Fa0/6 ↔ Fa0/16 simulando enlace trunk entre switches / Fa0/6 ↔ Fa0/16 simulating trunk link between switches

Initial Hypothesis	VLAN 20 no esta siendo permitida en el trunk link / VLAN 20 is not being allowed through the trunk link
Root Cause	Mismatch en allowed VLAN list: Fa0/6 permitia solo VLAN 10, Fa0/16 permitia 1-4094 / Allowed VLAN list mismatch: Fa0/6 only allowed VLAN 10, Fa0/16 allowed 1-4094
Command that revealed it	show interfaces trunk — muestra VLANs activas en el trunk / shows active VLANs on the trunk
Resolution	switchport trunk allowed vlan 10,20 en ambos puertos / on both ports
Status	CERRADO — VLAN 20 restaurada / CLOSED — VLAN 20 restored

PROCEDIMIENTO — 6 ETAPAS / PROCEDURE — 6 STAGES

#	ESPAÑOL	ENGLISH
1	ETAPA 1 — Baseline: show vlan brief, show interfaces status, show etherchannel summary	STAGE 1 — Baseline: show vlan brief, show interfaces status, show etherchannel summary
2	Confirmar estado inicial del switch antes de cualquier cambio	Confirm initial switch state before any change
3	ETAPA 2 — Creacion del escenario: conectar cable Fa0/6 ↔ Fa0/16	STAGE 2 — Scenario setup: connect cable Fa0/6 ↔ Fa0/16
4	Configurar ambos puertos en modo trunk: switchport mode trunk en Fa0/6 y Fa0/16	Configure both ports as trunk: switchport mode trunk on Fa0/6 and Fa0/16
5	Verificar trunk activo: show interfaces trunk — trunk formado correctamente entre puertos	Verify trunk active: show interfaces trunk — trunk formed correctly between ports
6	ETAPA 3 — Inducir falla: remover VLAN 20 de la allowed list de Fa0/6	STAGE 3 — Induce failure: remove VLAN 20 from Fa0/6 allowed list
7	Resultado: Fa0/6 permite solo VLAN 10 — Fa0/16 permite 1-4094 — MISMATCH detectado	Result: Fa0/6 allows only VLAN 10 — Fa0/16 allows 1-4094 — MISMATCH detected
8	Sintoma resultante: usuarios VLAN 20 sin conectividad, usuarios VLAN 10 normales	Resulting symptom: VLAN 20 users without connectivity, VLAN 10 users normal
9	ETAPA 4 — Diagnostico: ejecutar show interfaces trunk en ambos puertos	STAGE 4 — Diagnosis: run show interfaces trunk on both ports
10	Identificar mismatch: columna 'VLANs allowed and active in management domain' — VLAN 20 ausente en Fa0/6	Identify mismatch: column 'VLANs allowed and active in management domain' — VLAN 20 absent on Fa0/6
11	ETAPA 5 — Resolucion: switchport trunk allowed vlan 10,20 en Fa0/6 y Fa0/16	STAGE 5 — Resolution: switchport trunk allowed vlan 10,20 on Fa0/6 and Fa0/16
12	ETAPA 6 — Verificacion: show interfaces trunk → ambos puertos muestran VLANs 10,20 activas	STAGE 6 — Verification: show interfaces trunk → both ports show VLANs 10,20 active
13	Confirmar: show vlan brief → VLAN 10 y VLAN 20 activas. Ticket NOC-LAB-05 cerrado	Confirm: show vlan brief → VLAN 10 and VLAN 20 active. Ticket NOC-LAB-05 closed

ESTADO DEL TRUNK — ANTES Y DESPUES / TRUNK STATE — BEFORE AND AFTER

PUERTO / PORT	MODO / MODE	ANTES — VLANs permitidas	DESPUES — VLANs permitidas	ESTADO / STATUS
Fa0/6	trunk	10 (solo)	10, 20	✓ Correcto
Fa0/16	trunk	1-4094 (todas)	10, 20	✓ Correcto

REFERENCIA DE COMANDOS / COMMAND REFERENCE

COMANDO / COMMAND	SIGNIFICADO (ES)	MEANING (EN)
<code>show vlan brief</code>	Verifica VLANs activas y puertos asignados	Verifies active VLANs and assigned ports
<code>show interfaces status</code>	Muestra estado fisico de cada puerto	Shows physical status of each port
<code>show interfaces trunk</code>	COMANDO CLAVE — muestra modo trunk, VLANs permitidas, VLANs activas en el trunk	KEY COMMAND — shows trunk mode, allowed VLANs, active VLANs on the trunk
<code>show etherchannel summary</code>	Verifica bundles preexistentes (baseline)	Verifies pre-existing bundles (baseline)
<code>switchport mode trunk</code>	Configura el puerto en modo trunk	Sets port to trunk mode
<code>switchport trunk allowed vlan 10,20</code>	Permite solo VLAN 10 y 20 en el trunk	Allows only VLAN 10 and 20 on the trunk
<code>switchport trunk allowed vlan add 20</code>	Agrega VLAN 20 a la lista permitida sin borrar las existentes	Adds VLAN 20 to allowed list without removing existing ones

RESOLUCION PASO A PASO / STEP-BY-STEP RESOLUTION

#	COMANDO / COMMAND	ACCION (ES)	ACTION (EN)
1	<code>show interfaces trunk</code>	Identificar VLANs activas en el trunk — detectar mismatch	Identify active VLANs on trunk — detect mismatch
2	<code>interface FastEthernet 0/6</code>	Ingresa a configuracion de Fa0/6	Enter Fa0/6 configuration
3	<code>switchport trunk allowed vlan 10,20</code>	Permitir VLAN 10 y 20 en Fa0/6 (estaba solo VLAN 10)	Allow VLAN 10 and 20 on Fa0/6 (was only VLAN 10)
4	<code>interface FastEthernet 0/16</code>	Ingresa a configuracion de Fa0/16	Enter Fa0/16 configuration
5	<code>switchport trunk allowed vlan 10,20</code>	Permitir VLAN 10 y 20 en Fa0/16 (estaba 1-4094)	Allow VLAN 10 and 20 on Fa0/16 (was 1-4094)
6	<code>show interfaces trunk</code>	Verificar que ambos puertos muestran VLANs 10,20 activas	Verify both ports show VLANs 10,20 active
7	<code>show vlan brief</code>	Confirmar VLAN 10 y VLAN 20 activas en el switch	Confirm VLAN 10 and VLAN 20 active on switch

TROUBLESHOOTING DOCUMENTADO / DOCUMENTED TROUBLESHOOTING

HALLAZGO / FINDING	ACCION CORRECTIVA / CORRECTIVE ACTION
Usuarios VLAN 20 sin conectividad — VLAN 10 funcionando normalmente. / VLAN 20 users without connectivity — VLAN 10 working normally.	Diferencia de comportamiento entre VLANs apunta a problema de trunk allowed list, no a falla fisica. Diagnostico iniciado con show interfaces trunk. / Different behavior between VLANs points to trunk allowed list issue, not physical failure. Diagnosis started with show interfaces trunk.
show interfaces trunk revelo mismatch: Fa0/6 permitia solo VLAN 10, Fa0/16 permitia 1-4094. VLAN 20 no estaba en la interseccion. / show interfaces trunk revealed mismatch: Fa0/6 allowed only VLAN 10, Fa0/16 allowed 1-4094. VLAN 20 was not in the intersection.	Causa raiz identificada: allowed VLAN list inconsistente entre puertos del mismo trunk. Correccion: switchport trunk allowed vlan 10,20 en ambos puertos. / Root cause identified: inconsistent allowed VLAN list between trunk ports. Fix: switchport trunk allowed vlan 10,20 on both ports.
TRAMPA COMUN: usar 'switchport trunk allowed vlan 20' borra las VLANs previas y deja solo VLAN 20. / COMMON TRAP: using 'switchport trunk allowed vlan 20' removes previous VLANs and leaves only VLAN 20.	Usar 'switchport trunk allowed vlan add 20' para AGREGAR sin borrar, o especificar la lista completa: 'allowed vlan 10,20'. Verificar siempre con show interfaces trunk despues del cambio. / Use 'switchport trunk allowed vlan add 20' to ADD without removing, or specify full list: 'allowed vlan 10,20'. Always verify with show interfaces trunk after the change.
Este es uno de los problemas mas comunes en redes empresariales: trunk mal configurado genera perdida de conectividad silenciosa por VLAN. / This is one of the most common enterprise network issues: misconfigured trunk causes silent connectivity loss per VLAN.	En NOC, ante perdida de conectividad de un grupo especifico de usuarios, siempre verificar trunk allowed list antes de escalar. Resuelve el 60%+ de los casos similares. / In NOC, when a specific user group loses connectivity, always check trunk allowed list before escalating. Resolves 60%+ of similar cases.

RESULTADO / RESULT

RESULTADO (ES) VLAN trunk misconfiguration diagnosticada mediante show interfaces trunk. Mismatch en allowed VLAN list identificado y corregido. VLAN 20 restaurada. Ticket NOC-LAB-05 cerrado.	RESULT (EN) VLAN trunk misconfiguration diagnosed using show interfaces trunk. Allowed VLAN list mismatch identified and corrected. VLAN 20 restored. Ticket NOC-LAB-05 closed.
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KEY TAKEAWAY

"Cuando una VLAN especifica pierde conectividad y las demas funcionan bien, el primer comando es show interfaces trunk. La columna 'VLANs allowed and active' revela inmediatamente si hay un mismatch. En NOC esto es un ticket de 5 minutos si sabs donde mirar. La trampa clasica es usar 'allowed vlan X' en vez de 'allowed vlan add X' — el primero borra todo, el segundo agrega."

"When a specific VLAN loses connectivity and others work fine, the first command is show interfaces trunk. The 'VLANs allowed and active' column immediately reveals any mismatch. In NOC this is a 5-minute ticket if you know where to look. The classic trap is using 'allowed vlan X' instead of 'allowed vlan add X' — the first removes everything, the second adds."

NOC RELEVANCE — MINING / INDUSTRIAL ENVIRONMENT

In a mining environment (e.g., Newmont Cerro Negro), VLANs are used to segment OT (SCADA, PLCs, sensors) from IT (corporate PCs, IP phones) and from safety systems (emergency communications, gas detection). A misconfigured trunk allowed list can silently isolate an entire VLAN segment — for example, blocking SCADA traffic while corporate users remain online. The NOC engineer must identify per-VLAN impact patterns quickly, run `show interfaces trunk` as the first diagnostic step, apply targeted fixes without disrupting active VLANs, and document the change before and after with `show interfaces trunk` output in the ticket.